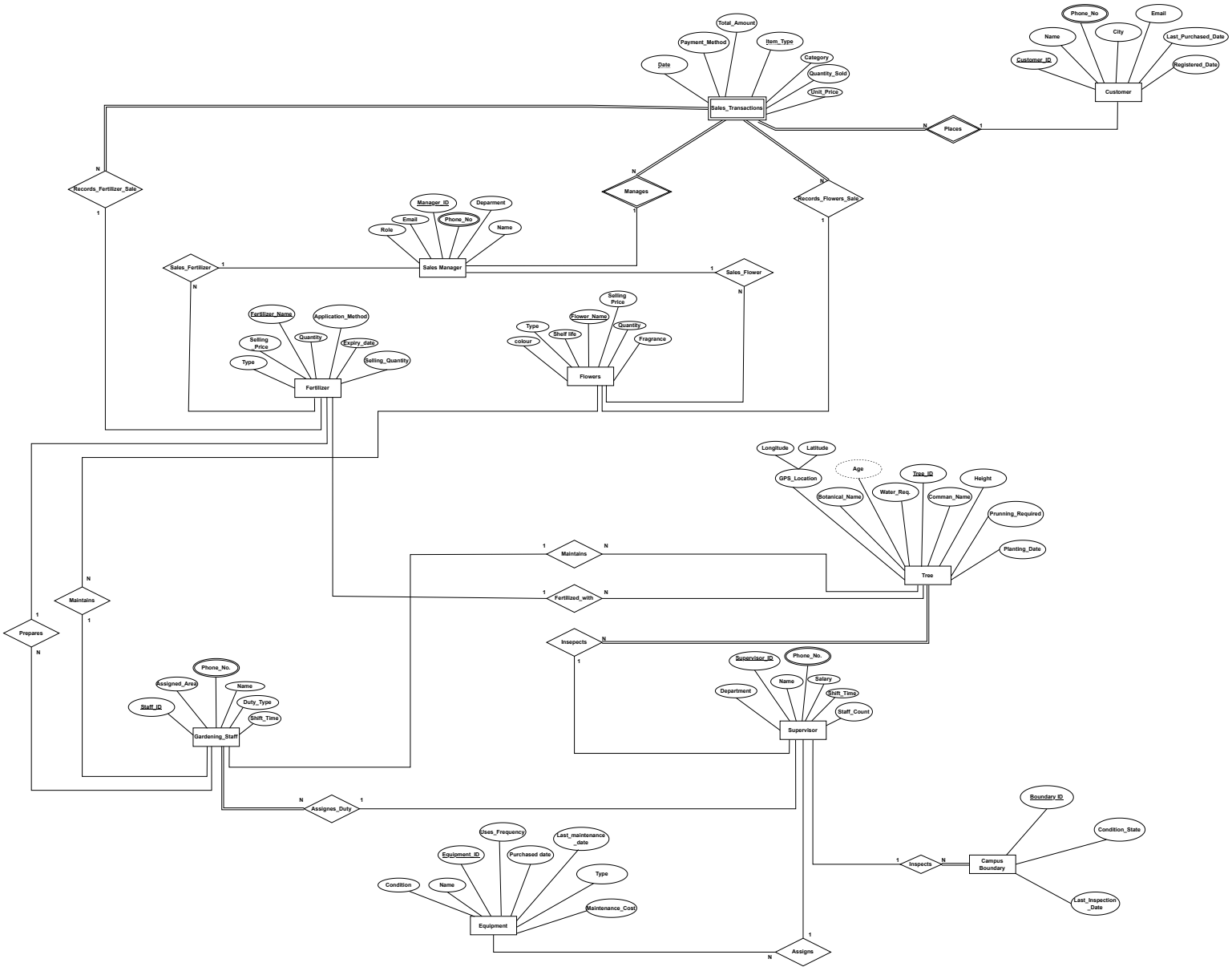
Group F4

DAIICT Green Campus Management

Vinas Moradiya - 202403060
Preet Hirapara - 202403013

FINAL ER Diagram V-3


Total_Amount
Payment_Method
Item_Type
Date
Category
Quantity_Sold
Unit_Price
Sales_Transactions
Phone_No
Email
Name
City
Last_Purchased_Date
Customer_ID
Registered_Date
Customer
Places
Records_Fertilizer_Sale
Manages
Records_Flowers_Sale
Manager_ID
Deparment
Email
Phone_No
Role
Name
Sales Manager
Sales_Fertilizer
Sales_Flower
Selling Price
Fertilizer_Name
Application_Method
Flower_Name
Type
Quantity
Selling _Price
Quantity
Shelf life
Expiry_date
colour
Fragrance
Type
Selling_Quantity
Fertilizer
Flowers
Longitude
Latitude
Age
Tree_ID
Height
GPS_Location
Water_Req.
Botanical_Name
Comman_Name
Prunning_Required
Planting_Date
Maintains
Tree
Maintains
Fertilized_with
Prepares
Insepects
Phone_No.
Supervisor_ID
Phone_No.
Assigned_Area
Name
Name
Salary
Department
Shift_Time
Staff_ID
Duty_Type
Staff_Count
Shift_Time
Supervisor
Gardening_Staff
Assignes_Duty
Boundary ID
Condition_State
Uses_Frequency
Last_maintenance _date
Equipment_ID
Purchased date
Inspects
Campus Boundary
Type
Condition
Name
Maintenance_Cost
Last_Inspection _Date
Equipment
Assigns

Group F4

DAIICT Green Campus Management

Vinas Moradiya - 202403060
Preet Hirapara - 202403013

Relational Model

Supervisor(Supervisor_ID, Name, Department, Shift_time, Staff_count, Salary)

Equipment(Equipment_ID, Supervisor_ID (FK), Gardening_Staff(FK), Name, Type, Purchased_Date, Condition, uses_frequency, Maintenance_cost, Last_maintanance_Date)

Inspects_campus_boundary(Boundary_ID, Supervisor_ID (FK), Condition_State, Last_inspection_Date)

Fertilizer(Fertilizer_Name, Manager_ID (FK), Type, Application_Method, Quantity, Selling_Quantity, Selling_price, Expiry_Date)

Gardening_Staff(Staff_ID,Supervisor_ID (FK) , Fertilizer_Name (FK), Name, Duty_Type, Assigned_Area, Shift_Time)

Tree(Tree_ID, Supervisor_ID (FK), Staff_ID (FK), Fertilizer_Name (FK), Common_Name, Botanical_Name, Age, Height, Longitude, Latitude, Planting_Date, Pruning_Required, Water_Req)

Sales_Manager(Manager_ID, Name, Email, Role, Department)

Flowers(Flower_Name, Gardenning_Staff (FK), Manager_ID(FK), Type, Colour, Fragrance, Shelf_Life, Quantity, Selling_Price)

Customer(Customer_ID, Name, Email, Registered_Date, Last_Purchased_Date, City)

Sales_Transaction(Date, Item_Type, Customer_ID (FK), Fertilizer_Name(FK), Flower_Name (FK), Manager_ID (FK), Category, Quantity_Sold, Unit_Price, Total_Amount, Payment_Method)

Customer(Customer_ID (FK), Phone_No)

Gardening(Staff_ID (FK), Phone_No)

Supervisor(Supervisor_ID (FK), Phone_No)

Sales_Manager(Manager_ID (FK), Phone_No)

Normalization & Schema Refinement

i. Identify and list all types of dependencies (PK, FK, Functional Dependencies) for each relation

Supervisor(Supervisor_ID, Name, Department, Shift_time, Staff_count, Salary)

> **PK** - Supervisor_ID
> **FK** -
> **Functional Dependencies** - Supervisor_ID → Name, Department, Shift_time, Staff_count, Salary

Equipment(Equipment_ID, Supervisor_ID (FK), Gardening_Staff (FK), Name, Type, Purchased_Date, Condition, uses_frequency, Maintenance_cost, Last_maintanance_Date)

> **PK** - Equipment_ID
> **FK** - Supervisor_ID, Gardening_Staff
> **Functional Dependencies** - Equipment_ID → Supervisor_ID, Gardening_Staff, Name, Type, Purchased_Date, Condition, uses_frequency, Maintenance_cost, Last_maintanance_Date

Inspects_campus_boundary(Boundary_ID, Supervisor_ID (FK), Condition_State, Last_inspection_Date)

PK - Boundary_ID
FK - Supervisor_ID
Functional Dependencies - Boundary_ID → Supervisor_ID, Condition_State, Last_inspection_Date

Fertilizer(Fertilizer_Name, Manager_ID (FK), Type, Application_Method, Quantity, Selling_Quantity, Selling_price, Expiry_Date)

PK - Fertilizer_Name
FK - Manager_ID
Functional Dependencies - Fertilizer_Name → Manager_ID, Type, Application_Method, Quantity, Selling_Quantity, Selling_price, Expiry_Date

Gardening_Staff(Staff_ID,Supervisor_ID (FK) , Fertilizer_Name (FK), Name, Duty_Type, Assigned_Area, Shift_Time)

PK - Staff_ID
FK - Supervisor_ID, Fertilizer_Name
Functional Dependencies - Staff_ID → Supervisor_ID, Fertilizer_Name, Name, Duty_Type, Assigned_Area, Shift_Time

Tree(Tree_ID, Supervisor_ID (FK), Staff_ID (FK), Fertilizer_Name (FK), Common_Name, Botanical_Name, Age, Height, Longitude, Latitude, Planting_Date, Pruning_Required, Water_Req)

PK - Tree_ID
FK - Supervisor_ID, Staff_ID, Fertilizer_Name
Functional Dependencies - Tree_ID → Supervisor_ID, Staff_ID, Fertilizer_Name, Common_Name, Botanical_Name, Age, Height, Longitude, Latitude, Planting_Date, Pruning_Required, Water_Req

Sales_Manager(Manager_ID, Name, Email, Role, Department)

PK - Manager_ID
FK -
Functional Dependencies - Manager_ID → Name, Email, Role, Department

Flowers(Flower_Name, Gardenning_Staff (FK), Manager_ID(FK), Type, Colour, Fragrance, Shelf_Life, Quantity, Selling_Price)

PK - Flower_Name
FK - Gardenning_Staff, Manager_ID
Functional Dependencies - Flower_Name → Gardenning_Staff, Type, Colour, Fragrance, Shelf_Life, Quantity, Selling_Price

Customer(Customer_ID, Name, Email, Registered_Date, Last_Purchased_Date, City)

PK - Customer_ID
FK -
Functional Dependencies - Customer_ID → Name, Email, Registered_Date, Last_Purchased_Date, City

Sales_Transaction(Date, Item_Type, Customer_ID (FK), Fertilizer_Name(FK), Flower_Name (FK), Manager_ID (FK), Category, Quantity_Sold, Unit_Price, Total_Amount, Payment_Method)

PK - Date, Item_Type, Customer_ID
FK - Customer_ID, Fertilizer_Name, Flower_Name, Manager_ID
Functional Dependencies - Transaction_ID, Customer_ID → Fertilizer_Name, Flower_Name, Manager_ID, Date, Category, Item_Type, Quantity_Sold, Unit_Price, Total_Amount, Payment_Method

Customer(Customer_ID (FK), Phone_No)

PK - Customer_ID, Phone_No
FK - Customer_ID

Gardening_Staff(Staff_ID (FK), Phone_No)

PK - Staff_ID , Phone_No
FK - Staff_ID

Supervisor(Supervisor_ID (FK), Phone_No)

PK - Supervisor_ID, Phone_No
FK - Supervisor_ID

Sales_Manager(Manager_ID (FK), Phone_No)

PK - Manager_ID, Phone_No
FK - Manager_ID

iii. Investigate every schema for the following

1. **List of redundancies existing for every schema which is part of the database**
2. **List of update, delete, and insert anomalies for every schema**

⇒ **Supervisor**(Supervisor_ID, Name, Department, Shift_time, Staff_count, Salary)

Redundancies

- Department and Shift_time values repeat for supervisors working in the same department.
- If department names change, the new value must be updated in many rows.

Anomalies

- **Update:** Changing department name or shift requires editing multiple rows results to inconsistency.
- **Delete:** Deleting the last supervisor of a department may erase department information.
- **Insert:** Cannot insert a new department until at least one supervisor is assigned.

⇒ **Equipment**(Equipment_ID, Supervisor_ID (FK), Gardening_Staff (FK), Name, Type, Purchased_Date, Condition, uses_frequency, Maintenance_cost, Last_maintanance_Date)

Redundancies

- Name, Type, Purchased_Date repeat for the same equipment assigned many times.

Anomalies

- **Update:** Changing equipment name or type requires many updates.
- **Delete:** Removing an assignment may delete history of equipment maintenance.
- **Insert:** Cannot record new assignment before equipment and staff exist.

⇒ **Inspects_campus_boundary**(Boundary_ID, Supervisor_ID (FK), Condition_State, Last_inspection_Date)

Redundancies

- Condition_State values (“Good”, “Needs Repair”) repeat.

Anomalies

- **Update:** If condition categories change, all rows must be updated.
- **Delete:** Deleting inspection removes record of boundary condition.
- **Insert:** Boundary section cannot exist before a supervisor is assigned.

⇒ **Fertilizer**(Fertilizer_Name, Manager_ID (FK), Type, Application_Method, Quantity, Selling_Quantity, Selling_price, Expiry_Date)

Redundancies

- Application_Method duplicated if multiple batches exist.

Anomalies

- **Update:** Price or method change results to multiple rows updated.
- **Delete:** Deleting last batch removes reference to fertilizer manager.
- **Insert:** Cannot insert new fertilizer without valid Manager_ID.

⇒ **Gardening_Staff**(Staff_ID, Supervisor_ID (FK), Fertilizer_Name (FK), Name, Duty_Type, Assigned_Area, Shift_Time)

Redundancies

- Same Shift_Time and Duty_Type may repeat across rows.

Anomalies

- **Update:** If supervisor changes, several staff records must be modified.
- **Delete:** Deleting a supervisor may orphan staff records.
- **Insert:** Cannot add staff before assigning valid supervisor/fertilizer.

⇒ **Tree**(Tree_ID, Supervisor_ID (FK), Staff_ID (FK), Fertilizer_Name (FK), Common_Name, Botanical_Name, Age, Height, Longitude, Latitude, Planting_Date, Pruning_Required, Water_Req)

Redundancies

- Common_Name and Botanical_Name repeated for all trees of the same species.

Anomalies

- **Update:** Changing species name requires updating multiple trees.
- **Delete:** Deleting staff or supervisor may orphan tree records.

- **Insert:** Cannot add tree until valid staff/supervisor/fertilizer exist.

⇒ **Sales_Manager**(Manager_ID, Name, Email, Role, Department)

Redundancies

- Department and Role values repeat if many managers share them.

Anomalies

- **Update:** Changing department name or role → multiple updates.
- **Delete:** Deleting a manager can orphan fertilizer, flower, or transaction records.
- **Insert:** Cannot assign fertilizers/flowers before manager exists.

⇒ **Flowers**(Flower_Name, Gardening_Staff (FK), Manager_ID (FK), Type, Colour, Fragrance, Shelf_Life, Quantity, Selling_Price)

Redundancies

- Selling_Price repeated for same flower varieties.

Anomalies

- **Update:** Price or colour change must be repeated in several rows.
- **Delete:** Deleting flower record removes details of responsible staff/manager.
- **Insert:** Cannot add flower until staff and manager exist.

⇒ **Customer**(Customer_ID, Name, Email, Registered_Date, Last_Purchased_Date, City)

Redundancies

- City repeated across many customers from the same location.

Anomalies

- **Update:** City or email change → single-row update (OK).

- **Delete:** Deleting customer removes all purchase history unless restricted.
- **Insert:** None major.

⇒ **Sales_Transaction**(Date, Item_Type, Customer_ID (FK), Fertilizer_Name(FK), Flower_Name (FK), Manager_ID (FK), Category, Quantity_Sold, Unit_Price, Total_Amount, Payment_Method)

Redundancies

- Unit_Price repeated for identical items.
- Total_Amount derivable = Quantity_Sold × Unit_Price → redundant.

Anomalies

- **Update:** If Unit_Price changes, old transactions become inconsistent.
- **Delete:** Removing a product or manager can orphan sales data.
- **Insert:** Cannot record sale before product and manager exist.

iv. Normalize the database up to 1NF (scalar values)

⇒ **Fertilizer**(Fertilizer_Name, Manager_ID (FK), Type, Application_Method, Quantity, Selling_Quantity, Selling_Price, Expiry_Date)

Application_Method could have multiple values (e.g., “spray/mix”) → multivalued.

Split into separate table for methods.

Solution :

Fertilizer(Fertilizer_Name, Manager_ID (FK), Type, Quantity, Selling_Quantity, Selling_price, Expiry_Date)
Fertilizer_Application(Fertilizer_Name (FK), Application_Method)

V. Normalize the database further to 2NF (Remove Partial Dependencies)

This relational model is already in 2NF as we don’t find any partial dependencies in any schema.

VI. Identify (and document) List of redundancies exiting for the schema in 2NF

⇒ **Supervisor**(Supervisor_ID, Name, Department, Shift_time, Staff_count, Salary)

Redundancies

- Department and Shift_time values repeat for supervisors working in the same department.
- If department names change, the new value must be updated in many rows.

Anomalies

- **Update:** Changing department name or shift requires editing multiple rows results to inconsistency.
- **Delete:** Deleting the last supervisor of a department may erase department information.
- **Insert:** Cannot insert a new department until at least one supervisor is assigned.

⇒ **Equipment**(Equipment_ID, Supervisor_ID (FK), Gardening_Staff (FK), Name, Type, Purchased_Date, Condition, uses_frequency, Maintenance_cost, Last_maintanance_Date)

Redundancies

- Name, Type, Purchased_Date repeat for the same equipment assigned many times.

Anomalies

- **Update:** Changing equipment name or type requires many updates.
- **Delete:** Removing an assignment may delete history of equipment maintenance.
- **Insert:** Cannot record new assignment before equipment and staff exist.

Inspects_campus_boundary(Boundary_ID, Supervisor_ID (FK), Condition_State, Last_inspection_Date)

Redundancies

- Condition_State values (“Good”, “Needs Repair”) repeat.

Anomalies

- **Update:** If condition categories change, all rows must be updated.
- **Delete:** Deleting inspection removes record of boundary condition.
- **Insert:** Boundary section cannot exist before a supervisor is assigned.

⇒ **Gardening_Staff**(Staff_ID, Supervisor_ID (FK), Fertilizer_Name (FK), Name, Duty_Type, Assigned_Area, Shift_Time)

Redundancies

- Same Shift_Time and Duty_Type may repeat across rows.

Anomalies

- **Update:** If supervisor changes, several staff records must be modified.
- **Delete:** Deleting a supervisor may orphan staff records.
- **Insert:** Cannot add staff before assigning valid supervisor/fertilizer.

⇒ **Tree**(Tree_ID, Supervisor_ID (FK), Staff_ID (FK), Fertilizer_Name (FK), Common_Name, Botanical_Name, Age, Height, Longitude, Latitude, Planting_Date, Pruning_Required, Water_Req)

Redundancies

- Common_Name and Botanical_Name repeated for all trees of the same species.

Anomalies

- **Update:** Changing species name requires updating multiple trees.
- **Delete:** Deleting staff or supervisor may orphan tree records.
- **Insert:** Cannot add tree until valid staff/supervisor/fertilizer exist.

⇒ **Sales_Manager**(Manager_ID, Name, Email, Role, Department)

Redundancies

- Department and Role values repeat if many managers share them.

Anomalies

- **Update:** Changing department name or role → multiple updates.
- **Delete:** Deleting a manager can orphan fertilizer, flower, or transaction records.

- **Insert:** Cannot assign fertilizers/flowers before manager exists.

⇒ **Flowers**(Flower_Name, Gardening_Staff (FK), Manager_ID (FK), Type, Colour, Fragrance, Shelf_Life, Quantity, Selling_Price)

Redundancies

- Selling_Price repeated for same flower varieties.

Anomalies

- **Update:** Price or colour change must be repeated in several rows.
- **Delete:** Deleting flower record removes details of responsible staff/manager.
- **Insert:** Cannot add flower until staff and manager exist.

⇒ **Customer**(Customer_ID, Name, Email, Registered_Date, Last_Purchased_Date, City)

Redundancies

- City repeated across many customers from the same location.

Anomalies

- **Update:** City or email change → single-row update (OK).
- **Delete:** Deleting customer removes all purchase history unless restricted.
- **Insert:** None major.

⇒ **Sales_Transaction**(Date, Item_Type, Customer_ID (FK), Fertilizer_Name(FK), Flower_Name (FK), Manager_ID (FK), Category, Quantity_Sold, Unit_Price, Total_Amount, Payment_Method)

Redundancies

- Unit_Price repeated for identical items.
- Total_Amount derivable = Quantity_Sold × Unit_Price → redundant.

Anomalies

- **Update:** If Unit_Price changes, old transactions become inconsistent.
- **Delete:** Removing a product or manager can orphan sales data.
- **Insert:** Cannot record sale before product and manager exist.

VII. Normalize it further to 3NF (Remove Transitive Dependencies)

⇒ **Supervisor**(Supervisor_ID, Name, Department, Shift_time, Staff_count, Salary)

> Staff_count depends on the Department (not directly on Supervisor_ID).
> → Different supervisors in the same department might share staff count.
>
> **Solution**:
>
> **Department_Info**(Department, Staff_count)
> **Supervisor**(Supervisor_ID, Name, Department (FK), Shift_time, Salary)

⇒ **Gardening_Staff**(Staff_ID,Supervisor_ID (FK) , Fertilizer_Name (FK), Name, Duty_Type, Assigned_Area, Shift_Time)

> Shift_Time might depend on Duty_Type (for example, all watering staff have same shift).
>
> **Solution:**
>
> **Duty_Info**(Duty_Type, Shift_Time)
> **Gardening_Staff**(Staff_ID,Supervisor_ID (FK) , Fertilizer_Name (FK), Name, Duty_Type (FK), Assigned_Area, Shift_Time)

⇒ **Tree**(Tree_ID, Supervisor_ID (FK), Staff_ID (FK), Fertilizer_Name (FK), Common_Name, Botanical_Name, Age, Height, Longitude, Latitude, Planting_Date, Pruning_Required, Water_Req)

Water_Req depends on Common_Name or Botanical_Name (same species needs same water).

Fertilizer_Name might depend on Common_Name (each species uses same fertilizer).

Solution:

Tree_Species(Common_Name, Botanical_Name, Water_Req, Fertilizer_Name (FK))
Tree(Tree_ID, Supervisor_ID (FK), Staff_ID (FK), Common_Name (FK), Age, Height, Longitude, Latitude, Planting_Date, Pruning_Required)

⇒ **Flowers**(Flower_Name, Gardenning_Staff (FK), Manager_ID(FK), Type, Colour, Fragrance, Shelf_Life, Quantity, Selling_Price)

Selling_Price depends on Type (all roses have same price).

Shelf_Life might depend on Type.

Solution:

Flower_Type(Type, Shelf_Life, Selling_Price)
Flowers(Flower_Name, Gardening_Staff (FK), Manager_ID(FK), Type (FK), Colour, Fragrance, Quantity)

⇒ **Sales_Manager**(Manager_ID, Name, Email, Role, Department)

Role may determine Department.

Solution :

Role_Info(Role, Department)
Sales_Manager(Manager_ID, Name, Email, Role (FK))

⇒ **Sales_Transaction**(Date, Item_Type, Customer_ID (FK), Fertilizer_Name(FK), Flower_Name (FK), Manager_ID (FK), Category, Quantity_Sold, Unit_Price, Total_Amount, Payment_Method)

Total_Amount depends on (Quantity_Sold, Unit_Price) not on the full key.

Category depends on Item_Type.

Solution :

Item_Details(Item_Type, Category, Unit_Price)
Sales_Transaction(Date, Item_Type (FK), Customer_ID (FK), Fertilizer_Name(FK), Flower_Name (FK), Manager_ID (FK), Quantity_Sold, Total_Amount, Payment_Method)

Final Relational Model After 3NF

Supervisor(Supervisor_ID, Name, Department (FK), Shift_time, Salary)

Department_Info(Department, Staff_count)

Equipment(Equipment_ID, Supervisor_ID (FK), Gardening_Staff(FK), Name, Type, Purchased_Date, Condition, uses_frequency, Maintenance_cost, Last_maintanance_Date)

Inspects_campus_boundary(Boundary_ID, Supervisor_ID (FK), Condition_State, Last_inspection_Date)

Fertilizer(Fertilizer_Name, Manager_ID (FK), Type, Quantity, Selling_Quantity, Selling_price, Expiry_Date)

Fertilizer_Application(Fertilizer_Name (FK), Application_Method)

Gardening_Staff(Staff_ID,Supervisor_ID (FK) , Fertilizer_Name (FK), Name, Duty_Type (FK), Assigned_Area, Shift_Time)

Duty_Info(Duty_Type, Shift_Time)

Tree(Tree_ID, Supervisor_ID (FK), Staff_ID (FK), Common_Name (FK), Age, Height, Longitude, Latitude, Planting_Date, Pruning_Required)

Tree_Species(Common_Name, Botanical_Name, Water_Req, Fertilizer_Name (FK))

Sales_Manager(Manager_ID, Name, Email, Role (FK))

Role_Info(Role, Department)

Flowers(Flower_Name, Gardening_Staff (FK), Manager_ID(FK), Type (FK), Colour, Fragrance, Quantity)

Flower_Type(Type, Shelf_Life, Selling_Price)

Customer(Customer_ID, Name, Email, Registered_Date, Last_Purchased_Date, City)

Sales_Transaction(Date, Item_Type (FK), Customer_ID (FK), Fertilizer_Name(FK), Flower_Name (FK), Manager_ID (FK), Quantity_Sold, Total_Amount, Payment_Method)

Item_Details(Item_Type, Category, Unit_Price)

Customer(Customer_ID (FK), Phone_No)

Gardening(Staff_ID (FK), Phone_No)

Supervisor(Supervisor_ID (FK), Phone_No)

Sales_Manager(Manager_ID (FK), Phone_No)